SUPPORTING INFORMATION

Phase Equilibria of CO_2-Water and CO_2-Brine at High Temperature: from Monte Carlo Simulations to Equation of State

Felipe Mourão Coelho,[†] Luís Fernando Mercier Franco,[*,†] and Abbas Firoozabadi[*,‡]

†*Universidade Estadual de Campinas (UNICAMP), Faculdade de Engenharia Química, Campinas-SP, 13083-852, Brazil*

‡*Rice University, Houston-TX, 77005, USA*

E-mail: lmfranco@unicamp.br; abbas.firoozabadi@rice.edu

Number of Pages: 31

Number of Figures: 16

Number of Tables: 3

CPA Equation of State

Compressibility Factor

The density of the mixture can be obtained from the compressibility factor, which can be derived from the residual Helmholtz free energy:

$$Z = 1 - \frac{v}{RT}\left(\frac{\partial A^r}{\partial V}\right)_{T,N_i}. \tag{S1}$$

The compressibility factor can be divided into terms representing different interactions. We also add a term related to the change in the relative permittivity (perm), which is from the DH and Born term (more details on the origin of this term are presented afterward):

$$Z = Z^{\text{phys}} + Z^{\text{assoc}} + Z^{\text{DH}} + Z^{\text{Born}} + Z^{\text{Perm}}. \tag{S2}$$

The compressibility factor expression for the physical term is given by:

$$Z^{\text{phys}} = \frac{Z}{Z-B} - \frac{A}{Z+B}, \tag{S3}$$

where A and B are the dimensionless SRK parameters:

$$A = \frac{aP}{R^2T^2}, \tag{S4}$$

$$B = \frac{bP}{RT}. \tag{S5}$$

The association expression for the compressibility factor is given by:

$$Z^{\text{assoc}} = \frac{1}{2}\left(1 + \frac{\eta}{g_\eta}\frac{\mathrm{d}g_\eta}{\mathrm{d}\eta}\right)\sum_i x_i \sum_{A\in i}\left(\chi_{A_i} - 1\right). \tag{S6}$$

In the DH and Born terms, we split them into a contribution at fixed relative permittivity and varying relative permittivity, which originates Z^{Perm}:

$$-\frac{v}{RT}\left(\frac{\partial A^{\mathrm{DH}}}{\partial V}\right)_{T,N_i} = -\frac{v}{RT}\left(\frac{\partial A^{\mathrm{DH}}}{\partial V}\right)_{T,N_i,\epsilon_r} - \frac{v}{RT}\left(\frac{\partial A^{\mathrm{DH}}}{\partial \epsilon_r}\right)_{T,V,N_i}\left(\frac{\partial \epsilon_r}{\partial V}\right)_{T,N_i}, \tag{S7}$$

$$Z^{\mathrm{DH}} = -\frac{v}{RT}\left(\frac{\partial A^{\mathrm{DH}}}{\partial V}\right)_{T,N_i,\epsilon_r} = \frac{v}{4\pi N_A \sum_j x_j z_j^2}\sum_j x_j z_j^2\left[X_j - \frac{\kappa^3}{2\left(1+\kappa\sigma_j\right)}\right], \tag{S8}$$

$$-\frac{v}{RT}\left(\frac{\partial A^{\mathrm{Born}}}{\partial V}\right)_{T,N_i} = -\frac{v}{RT}\left(\frac{\partial A^{\mathrm{Born}}}{\partial V}\right)_{T,N_i,\epsilon_r} - \frac{v}{RT}\left(\frac{\partial A^{\mathrm{Born}}}{\partial \epsilon_r}\right)_{T,V,N_i}\left(\frac{\partial \epsilon_r}{\partial V}\right)_{T,N_i}, \tag{S9}$$

$$Z^{\mathrm{Born}} = -\frac{v}{RT}\left(\frac{\partial A^{\mathrm{Born}}}{\partial V}\right)_{T,N_i,\epsilon_r} = 0, \tag{S10}$$

$$Z^{\mathrm{Perm}} = -\frac{v}{RT}\left[\left(\frac{\partial A^{\mathrm{DH}}}{\partial \epsilon_r}\right)_{T,V,N_i} + \left(\frac{\partial A^{\mathrm{Born}}}{\partial \epsilon_r}\right)_{T,V,N_i}\right]\left(\frac{\partial \epsilon_r}{\partial V}\right)_{T,N_i}, \tag{S11}$$

$$Z^{\mathrm{Perm}} = -\left[\frac{\kappa^2}{8\pi N_A c\epsilon_r \sum_j x_j z_j^2}\sum_j\left(\frac{\kappa x_j z_j^2}{1+\kappa\sigma_j} - \frac{x_j z_j^2}{R_{B,j}}\right)\right] V\left(\frac{\partial \epsilon_r}{\partial V}\right)_{T,N_i}. \tag{S12}$$

We, therefore, need an expression for the variation of the dielectric constant with volume.

Fugacity Coefficient

When two phases are in equilibrium, temperature, pressure, and fugacity of each component in both phases are equal. In terms of chemical equilibrium, the fugacity equality of component i between phases I and II is given by:

$$x_i^I \phi_i^I = x_i^{II} \phi_i^{II}, \tag{S13}$$

where ϕ_i is the fugacity coefficient of component i, which can be derived from the residual Helmholtz free energy as:

$$\ln \phi_i = \frac{1}{RT}\left(\frac{\partial A^r}{\partial N_i}\right)_{T,V,N_j} - \ln Z. \tag{S14}$$

Similarly to the compressibility factor, the fugacity coefficient can be divided into terms representing each one of the interactions:

$$\ln \phi_i = \ln \phi_i^{\text{phys}} + \ln \phi_i^{\text{assoc}} + \ln \phi_i^{\text{DH}} + \ln \phi_i^{\text{Born}} + \ln \phi_i^{\text{Perm}}. \tag{S15}$$

Separating the relative permittivity contribution as done in the compressibility factor, the expressions for the physical, association, DH and Born and relative permittivity fugacity coefficients are given, respectively, by:

$$\begin{aligned}\ln \phi_i^{\text{phys}} &= -\ln\left(Z-B\right) + \frac{B_i}{B}\left(\frac{B}{Z-B} - \frac{A}{Z+B}\right) \\ &\quad - \left\{\frac{A_i}{B_i} + \frac{1}{B\ln 2}\left[B_i\frac{g^{E\infty}}{RT} - \sum_j x_j\left(B_i\frac{\Delta U_{ij}}{RT} + B_j\frac{\Delta U_{ji}}{RT}\right)\right]\right\}\ln\left(1+\frac{B}{Z}\right),\end{aligned} \tag{S16}$$

$$\ln \phi_i^{\text{assoc}} = \sum_{A\in i}\eta_{A_i}\ln\chi_{A_i} + \frac{B_i}{8g_\eta Z}\frac{\mathrm{d}g_\eta}{\mathrm{d}\eta}\sum_j x_j \sum_{A\in j}\left(\chi_{A_j}-1\right), \tag{S17}$$

$$\ln \phi_i^{\text{DH}} = \frac{z_i^2 v}{4\pi N_A \sum_j x_j z_j^2}\left[\frac{\sum_j x_j z_j^2 X_j}{\sum_j x_j z_j^2} - X_i - \frac{\kappa^3}{2\sum_j x_j z_j^2}\sum_j\left(\frac{x_j z_j^2}{1+\kappa\sigma_j}\right)\right], \tag{S18}$$

$$\ln \phi_i^{\text{Born}} = \frac{N_A e^2 z_i^2}{8\pi\epsilon_0 RT R_{B,i}}\left(\frac{1}{\epsilon_r} - 1\right), \tag{S19}$$

$$\ln \phi_i^{\text{Perm}} = \left[\frac{\kappa^2}{8\pi N_A c\epsilon_r \sum_j x_j z_j^2}\sum_j\left(\frac{\kappa x_j z_j^2}{1+\kappa\sigma_j} - \frac{x_j z_j^2}{R_{B,j}}\right)\right] N\left(\frac{\partial\epsilon_r}{\partial N_i}\right)_{T,V,N_j}. \tag{S20}$$

Here, we need an expression for the variation of the dielectric constant with composition.

Relative Permitivitty

To determine the relative permittivity contribution to the compressibility factor (Eq. S12) and to the fugacity coefficient (Eq. S20), one needs to evaluate the derivatives $\partial\epsilon_r/\partial V$ and $\partial\epsilon_r/\partial N_i$. We show how to obtain these derivatives from eCPA Eos and the relative permittivity primitive model from **?**.[?] Assuming water as the only polar molecule in the mixture with a net dipole moment, we can define function $F\left(\epsilon_r, \epsilon_\infty, N_i, V\right)$:

$$F\left(\epsilon_r, \epsilon_\infty, N_i, V\right) = \frac{\left(2\epsilon_r + \epsilon_\infty\right)\left(\epsilon_r - \epsilon_\infty\right)}{\epsilon_r\left(\epsilon_\infty + 2\right)^2} - \frac{N_A}{9\epsilon_0 k_B T v} x_w g_w \mu_{0w}^2 = 0. \tag{S21}$$

The water Kirkwood g-factor and the probability of association between water and other molecules are given by:?

$$g_w = 1 + \frac{z_{ww}\mathcal{P}_{ww}\cos\gamma_{ww}}{\mathcal{P}_w\cos\theta_{ww} + 1}, \tag{S22}$$

$$\mathcal{P}_w = \sum_j \mathcal{P}_{wj}, \tag{S23}$$

$$\mathcal{P}_{wj} = \eta_{A_w} c x_j \chi_{A_w}\chi_{B_j}\delta_{A_i B_j}, \tag{S24}$$

where η_{A_w} is the number of associating sites type A in water (2 for the 4C scheme).

To obtain the derivatives $\partial\epsilon_r/\partial V$ and $\partial\epsilon_r/\partial N_i$, we differentiate the function F:

$$\mathrm{d}F = \left(\frac{\partial F}{\partial\epsilon_r}\right)\mathrm{d}\epsilon_r + \left(\frac{\partial F}{\partial\epsilon_\infty}\right)\mathrm{d}\epsilon_\infty + \left(\frac{\partial F}{\partial V}\right)\mathrm{d}V + \sum_i\left(\frac{\partial F}{\partial N_i}\right)\mathrm{d}N_i = 0. \tag{S25}$$

To simplify the notation, we omit each term that is kept constant at each partial derivative. The dielectric constant derivatives can be expressed as:

$$V\left(\frac{\partial\epsilon_r}{\partial V}\right)_{T,Ni} = -\left(\frac{\partial F}{\partial\epsilon_r}\right)^{-1}\left[\left(\frac{\partial F}{\partial\epsilon_\infty}\right)V\left(\frac{\partial\epsilon_\infty}{\partial V}\right) + V\left(\frac{\partial F}{\partial V}\right)\right], \tag{S26}$$

$$N\left(\frac{\partial\varepsilon_r}{\partial N_i}\right)_{T,V,Nj} = -\left(\frac{\partial F}{\partial\varepsilon_r}\right)^{-1}\left[\left(\frac{\partial F}{\partial\varepsilon_\infty}\right)N\left(\frac{\partial\varepsilon_\infty}{\partial N_i}\right) + N\left(\frac{\partial F}{\partial N_i}\right)\right]. \tag{S27}$$

Each term of Eqs. (S26) and (S27) is given by the following expressions:

$$\left(\frac{\partial F}{\partial\epsilon_r}\right) = \frac{2\epsilon_r^2 + \epsilon_\infty^2}{\epsilon_r^2\left(\epsilon_\infty + 2\right)^2}, \tag{S28}$$

$$\left(\frac{\partial F}{\partial\epsilon_\infty}\right) = \frac{\epsilon_r\epsilon_\infty - 4\epsilon_\infty - 4\epsilon_r^2 - 2\epsilon_r}{\epsilon_r\left(\epsilon_\infty + 2\right)^3}, \tag{S29}$$

$$V\left(\frac{\partial \epsilon_\infty}{\partial V}\right) = -\frac{3M_{\epsilon_\infty}}{\left(1-M_{\epsilon_\infty}\right)^2}, \tag{S30}$$

$$N\left(\frac{\partial \epsilon_\infty}{\partial N_i}\right) = \frac{N_A c \alpha_{0i}}{\epsilon_0\left(1-M_{\epsilon_\infty}\right)^2}, \tag{S31}$$

$$V\left(\frac{\partial F}{\partial V}\right) = V\left(\frac{\partial F}{\partial V}\right)_{g_w} + \left(\frac{\partial F}{\partial g_w}\right) V\left(\frac{\partial g_w}{\partial V}\right) = \frac{N_A c x_w \mu_{0w}^2}{9\epsilon_0 k_B T}\left[g_w - V\left(\frac{\partial g_w}{\partial V}\right)\right], \tag{S32}$$

$$N\left(\frac{\partial F}{\partial N_i}\right) = N\left(\frac{\partial F}{\partial N_i}\right)_{g_w} + \left(\frac{\partial F}{\partial g_w}\right) N\left(\frac{\partial g_w}{\partial N_i}\right) = -\frac{N_A c x_w \mu_{0w}^2}{9\varepsilon_0 k_B T}\left[\frac{g_w \Gamma_{iw}}{x_w} + N\left(\frac{\partial g_w}{\partial N_i}\right)\right], \tag{S33}$$

where $\Gamma_{ij} = 1$ if $i = j$, and $\Gamma_{ij} = 0$ otherwise, and

$$M_{\epsilon_\infty} = \frac{\epsilon_\infty - 1}{\epsilon_\infty + 2}. \tag{S34}$$

Now we evaluate two special cases. In the first one, water is the only associating molecule and we can obtain an analytical expression for $\partial g_w/\partial V$ and $\partial g_w/\partial N_i$. In the second case, besides water self-association, we also have CO_2-H_2O cross-association, in such a way that numerical methods are used to obtain the derivatives of $\partial \chi_w/\partial V$ and $\partial \chi_w/\partial N_i$

Special Case I: Water as single associating molecule

Assuming that $\varphi = V$, N or N if $\Phi = V$, N_w or N_c.

$$\varphi\left(\frac{\partial g_w}{\partial \Phi}\right) = \left(\frac{\partial g_w}{\partial \mathcal{P}_{ww}}\right)\left(\frac{\partial \mathcal{P}_{ww}}{\partial \chi_w}\right)\varphi\left(\frac{\partial \chi_w}{\partial \Phi}\right), \tag{S35}$$

$$\mathcal{P}_w = \mathcal{P}_{ww} = 1 - \chi_w, \tag{S36}$$

$$\left(\frac{\partial g_w}{\partial \mathcal{P}_{ww}}\right) = \frac{g_w - 1}{\mathcal{P}_{ww}\left(\mathcal{P}_{ww}\cos\theta_{ww} + 1\right)}, \tag{S37}$$

$$\left(\frac{\partial \mathcal{P}_{ww}}{\partial \chi_w}\right) = -1, \tag{S38}$$

$$V\left(\frac{\partial \chi_w}{\partial V}\right) = \frac{1}{4cx_w\delta}\left(\frac{1+4cx_w\delta}{\sqrt{1+8cx_w\delta}} - 1\right)\left(1 + \frac{\eta}{g_\eta}\frac{\mathrm{d}g_\eta}{\mathrm{d}\eta}\right), \tag{S39}$$

$$N\left(\frac{\partial \chi_w}{\partial N_i}\right) = \frac{1}{4cx_w\delta}\left(1 - \frac{1+4cx_w\delta}{\sqrt{1+8cx_w\delta}}\right)\left(\frac{\Gamma_{iw}}{x_w} + \frac{cb_i}{4g_\eta}\frac{\mathrm{d}g_\eta}{\mathrm{d}\eta}\right). \tag{S40}$$

Special Case II: Water self-association + $\mathbf{CO_2}$ cross-association

$$\mathcal{P}_w = \mathcal{P}_{ww} + \mathcal{P}_{wc}, \tag{S41}$$

$$\mathcal{P}_{ww} = 2cx_w\chi_w^2\delta_w, \tag{S42}$$

$$\mathcal{P}_{wc} = 2cx_c\chi_w\chi_c S_{cw}\delta_w, \tag{S43}$$

$$\begin{aligned}\varphi\left(\frac{\partial g_w}{\partial \Phi}\right) &= \left(\frac{\partial g_w}{\partial \mathcal{P}_{ww}}\right)\varphi\left(\frac{\partial \mathcal{P}_{ww}}{\partial \Phi}\right) + \left(\frac{\partial g_w}{\partial \mathcal{P}_w}\right)\varphi\left(\frac{\partial \mathcal{P}_w}{\partial \Phi}\right)\\ &= \frac{g_w-1}{\mathcal{P}_{ww}}\varphi\left(\frac{\partial \mathcal{P}_{ww}}{\partial \Phi}\right) + \frac{(g_w-1)\cos\theta_{ww}}{\mathcal{P}_w\cos\theta_{ww}+1}\left[\varphi\left(\frac{\partial \mathcal{P}_{ww}}{\partial \Phi}\right) + \varphi\left(\frac{\partial \mathcal{P}_{wc}}{\partial \Phi}\right)\right].\end{aligned} \tag{S44}$$

$\mathcal{P}_{ww} = f(N_w, V, \delta_w, \chi_w)$, whereas $\mathcal{P}_{wc} = f(N_c, V, \delta_w, \chi_w, \chi_c,)$. Their derivatives with respect of V and N_i are then given, respectively, by:

$$V\frac{\partial \mathcal{P}_{ww}}{\partial V} = \mathcal{P}_{ww}\left[-1 + \frac{\delta_w\eta}{g_\eta}\frac{\mathrm{d}g_\eta}{\mathrm{d}\eta} + \frac{2}{\chi_w}V\left(\frac{\partial \chi_w}{\partial V}\right)\right], \tag{S45}$$

$$V\frac{\partial \mathcal{P}_{wc}}{\partial V} = \mathcal{P}_{wc}\left[-1 + \frac{\delta_w\eta}{g_\eta}\frac{\mathrm{d}g_\eta}{\mathrm{d}\eta} + \frac{1}{\chi_w}V\left(\frac{\partial \chi_w}{\partial N_i}\right) + \frac{1}{\chi_c}V\left(\frac{\partial \chi_c}{\partial N_i}\right)\right], \tag{S46}$$

$$N\frac{\partial \mathcal{P}_{ww}}{\partial N_i} = \mathcal{P}_{ww}\left[\frac{\Gamma_{iw}}{x_w} + \frac{\delta_w\eta b_i}{g_\eta b}\frac{\mathrm{d}g_\eta}{\mathrm{d}\eta} + \frac{2}{\chi_w}N\left(\frac{\partial \chi_w}{\partial N_i}\right)\right], \tag{S47}$$

$$N\frac{\partial \mathcal{P}_{wc}}{\partial N_i} = \mathcal{P}_{wc}\left[\frac{\Gamma_{ic}}{x_c} + \frac{\delta_w\eta b_i}{g_\eta b}\frac{\mathrm{d}g_\eta}{\mathrm{d}\eta} + \frac{1}{\chi_w}N\left(\frac{\partial \chi_w}{\partial N_i}\right) + \frac{1}{\chi_c}N\left(\frac{\partial \chi_c}{\partial N_i}\right)\right]. \tag{S48}$$

The derivatives of χ_i with respect to V and $N_i$are obtained numerically. First, we define two functions G and H:

$$G(\chi_w, \chi_c, \delta_w, V, N_w) = \chi_c - \frac{V}{V + 2N_w\chi_w S_{cw}\delta_w} = 0, \tag{S49}$$

$$H(\chi_w, \chi_c, \delta_w, V, N_w, N_c) = \chi_w - \frac{V}{V + 2N_w\chi_w\delta_w + 2N_c\chi_c S_{cw}\delta_w} = 0, \tag{S50}$$

$$\mathrm{d}G = \left(\frac{\partial G}{\partial \chi_w}\right)\mathrm{d}\chi_w + \left(\frac{\partial G}{\partial \chi_c}\right)\mathrm{d}\chi_c + \left(\frac{\partial G}{\partial \delta_w}\right)\mathrm{d}\delta_w + \left(\frac{\partial G}{\partial V}\right)\mathrm{d}V + \left(\frac{\partial G}{\partial N_w}\right)\mathrm{d}N_w = 0, \tag{S51}$$

$$\mathrm{d}H = \left(\frac{\partial H}{\partial \chi_w}\right)\mathrm{d}\chi_w + \left(\frac{\partial H}{\partial \chi_c}\right)\mathrm{d}\chi_c + \left(\frac{\partial H}{\partial \delta_w}\right)\mathrm{d}\delta_w + \left(\frac{\partial H}{\partial V}\right)\mathrm{d}V + \left(\frac{\partial H}{\partial N_w}\right)\mathrm{d}N_w + \left(\frac{\partial H}{\partial N_c}\right)\mathrm{d}N_c = 0, \tag{S52}$$

$$\varphi\left(\frac{\partial \chi_c}{\partial \Phi}\right) = -\left(\frac{\partial G}{\partial \chi_c}\right)^{-1}\left[\left(\frac{\partial G}{\partial \chi_w}\right)\varphi\left(\frac{\partial \chi_w}{\partial \Phi}\right) + \left(\frac{\partial G}{\partial \delta_w}\right)\varphi\left(\frac{\partial \delta_w}{\partial \Phi}\right) + \varphi\left(\frac{\partial G}{\partial \Phi}\right)\right], \tag{S53}$$

$$\varphi\left(\frac{\partial \chi_w}{\partial \Phi}\right) = -\left(\frac{\partial H}{\partial \chi_w}\right)^{-1}\left[\left(\frac{\partial H}{\partial \chi_c}\right)\varphi\left(\frac{\partial \chi_c}{\partial \Phi}\right) + \left(\frac{\partial H}{\partial \delta_w}\right)\varphi\left(\frac{\partial \delta_w}{\partial \Phi}\right) + \varphi\left(\frac{\partial H}{\partial \Phi}\right)\right]. \tag{S54}$$

We have three new unknowns: $\partial\chi_w/\partial V, \partial\chi_w/\partial N_w$, and $\partial\chi_w/\partial N_c$ that can be solved numerically by coupling Eqs. S53 and S54, knowing that $\partial\chi_c/\partial N_c = 0$. All the other derivatives are given next:

$$\left(\frac{\partial G}{\partial \chi_w}\right) = 2x_w c S_{cw}\delta_w\chi_w^2, \tag{S55}$$

$$\left(\frac{\partial G}{\partial \chi_c}\right) = 1, \tag{S56}$$

$$\left(\frac{\partial G}{\partial \delta_w}\right) = -\chi_c^2\left[1 + 2x_w c S_{cw}\chi_w\left(\delta_w - 1\right)\right], \tag{S57}$$

$$\left(\frac{\partial H}{\partial \chi_w}\right) = 1 + 2x_w c\delta_w\chi_w^2, \tag{S58}$$

$$\left(\frac{\partial H}{\partial \chi_c}\right) = 2x_c c S_{cw}\delta_w\chi_c^2, \tag{S59}$$

$$\left(\frac{\partial H}{\partial \delta_w}\right) = -\chi_w^2\left[1 + \left(2x_w c\chi_w + 2x_c c S_{cw}\chi_w\right)\left(\delta_w - 1\right)\right], \tag{S60}$$

$$V\left(\frac{\partial G}{\partial V}\right) = -2x_w c S_{cw}\delta_w\chi_w^3, \tag{S61}$$

$$V\left(\frac{\partial H}{\partial V}\right) = -\chi_w^2\left(2x_w c\delta_w\chi_w + 2x_c c S_{cw}\delta_w\chi_c\right), \tag{S62}$$

$$V\left(\frac{\partial \delta_w}{\partial V}\right) = -\frac{\delta_w\eta}{g_\eta}\frac{\mathrm{d}g_\eta}{\mathrm{d}\eta}, \tag{S63}$$

$$N\left(\frac{\partial G}{\partial N_w}\right) = 2c S_{cw}\delta_w\chi_w^3, \tag{S64}$$

$$N\left(\frac{\partial H}{\partial N_w}\right) = 2c\delta_w\chi_w^3, \tag{S65}$$

$$N\left(\frac{\partial H}{\partial N_c}\right) = 2cS_{cw}\delta_w\chi_w^2\chi_c, \tag{S66}$$

$$N\left(\frac{\partial \delta_w}{\partial N_i}\right) = \frac{\delta_w\eta b_i}{g_\eta b}\frac{\mathrm{d}g_\eta}{\mathrm{d}\eta}. \tag{S67}$$

Newton Solver for the Aqueous Inner Loop

Computing $\ln\phi_i$ in the aqueous phase requires the simultaneous solution of three self-consistency equations in the unknowns $\mathbf{v} = (Z_w,\, \varepsilon_r,\, \chi_w)^\top$:

$$F_0(Z_w, \varepsilon_r, \chi_w) = Z_w - Z_w^{\text{phys}} - Z_w^{\text{assoc}} - Z_w^{\text{DH}} - Z_w^{\text{Perm}} = 0, \tag{S68}$$

$$F_1(Z_w, \varepsilon_r, \chi_w) = T_1(\varepsilon_r, \varepsilon_\infty) - T_2(Z_w, \chi_w) = 0, \tag{S69}$$

$$F_2(Z_w, \varepsilon_r, \chi_w) = \chi_w - \chi_w^{\text{new}}(Z_w, \chi_w) = 0. \tag{S70}$$

Here Z_w^{phys}, Z_w^{assoc}, Z_w^{DH} and Z_w^{Perm} are defined in Eqs. (S3)–(S12), T_1 and T_2 are, respectively, the left- and right-hand sides of the relative permittivity self-consistency equation (S21), and χ_w^{new} follows from the H-function, Eq. (S50).

Writing the dimensionless association strength $\Delta = \delta_w P/(RT)$, the cross-associating fraction is

$$\chi_c = \frac{Z_w}{Z_w + 2x_w\chi_w S_{cw}\Delta}, \qquad D_c \equiv Z_w + 2x_w\chi_w S_{cw}\Delta, \tag{S71}$$

and the aqueous H_2O fraction reads

$$\chi_w^{\text{new}} = \frac{Z_w}{Z_w + 2x_w\chi_w\Delta + 2x_c\chi_c S_{cw}\Delta}, \qquad D_w \equiv Z_w + 2x_w\chi_w\Delta + 2x_c\chi_c S_{cw}\Delta. \tag{S72}$$

Newton's method is applied to Eqs. (S68)–(S70):

$$\mathbf{v}_{k+1} = \mathbf{v}_k - \mathbf{J}(\mathbf{v}_k)^{-1}\,\mathbf{F}(\mathbf{v}_k), \tag{S73}$$

where the 3×3 analytical Jacobian $J_{ij} = \partial F_i/\partial v_j$ is derived below.

Row 2: $\partial F_2/\partial \mathbf{v}$

Because $F_2 = \chi_w - Z_w/D_w$, the chain of dependencies $\chi_c(\chi_w)$ and $\chi_c(Z_w)$ must be carried through D_w.

The partial of χ_c with respect to Z_w is

$$\frac{\partial \chi_c}{\partial Z_w} = \frac{D_c - Z_w\,\partial D_c/\partial Z_w}{D_c^2}, \qquad \frac{\partial D_c}{\partial Z_w} = 1 + 2x_w\chi_w S_{cw}\frac{\partial \Delta}{\partial Z_w}, \tag{S74}$$

with $\partial\Delta/\partial Z_w$ obtained from $\Delta = \delta_w P/(RT)$ and $\rho \propto Z_w^{-1}$ at fixed T, P:

$$Z_w\frac{\partial \Delta}{\partial Z_w} = Z_w\frac{\partial \delta_w}{\partial Z_w}\frac{P}{RT} = V\frac{\partial \delta_w}{\partial V}\frac{P}{RT} = -\frac{\Delta\eta}{g_\eta}\frac{\mathrm{d}g_\eta}{\mathrm{d}\eta}. \tag{S75}$$

Analogously, $\partial\chi_c/\partial\chi_w = -Z_w \cdot 2x_w S_{cw}\Delta/D_c^2$. With D_w defined in Eq. (S72), the Jacobian entries are:

$$J_{20} = -\frac{\partial \chi_w^{\text{new}}}{\partial Z_w} = -\frac{D_w - Z_w\,\partial D_w/\partial Z_w}{D_w^2}, \tag{S76}$$

$$J_{21} = 0, \tag{S77}$$

$$J_{22} = 1 - \frac{\partial \chi_w^{\text{new}}}{\partial \chi_w} = 1 + \frac{Z_w\,\partial D_w/\partial \chi_w}{D_w^2}, \tag{S78}$$

where

$$\frac{\partial D_w}{\partial Z_w} = 1 + (2x_w\chi_w + 2x_c\chi_c S_{cw})\,\frac{\partial \Delta}{\partial Z_w} + 2x_c S_{cw}\Delta\,\frac{\partial \chi_c}{\partial Z_w}, \tag{S79}$$

$$\frac{\partial D_w}{\partial \chi_w} = 2x_w\Delta + 2x_c S_{cw}\Delta\,\frac{\partial \chi_c}{\partial \chi_w}. \tag{S80}$$

Row 1: $\partial F_1/\partial \mathbf{v}$

$T_1 = F(\varepsilon_r, \varepsilon_\infty)$ is the left-hand side of Eq. (S21). Its partial derivatives are those already given in Eqs. (S28) and (S29). $T_2 = (N_A c/9\varepsilon_0 k_B T)\, x_w g_w \mu_{0w}^2$ depends on Z_w through $\rho = P/(Z_w RT)$ and on χ_w through g_w.

The three entries are:

$$J_{10} = \left(\frac{\partial F}{\partial \varepsilon_\infty}\right) \frac{V(\partial \varepsilon_\infty/\partial V)}{Z_w} - \frac{N_A \mu_{0w}^2}{9\varepsilon_0 k_B T}\, x_w \left(\frac{\partial \rho}{\partial Z_w}\, g_w + \rho\, \frac{\partial g_w}{\partial Z_w}\right), \tag{S81}$$

$$J_{11} = \left(\frac{\partial F}{\partial \varepsilon_r}\right) = \frac{2\varepsilon_r^2 + \varepsilon_\infty^2}{\varepsilon_r^2(\varepsilon_\infty + 2)^2}, \tag{S82}$$

$$J_{12} = -\frac{N_A \rho \mu_{0w}^2}{9\varepsilon_0 k_B T}\, x_w\, \frac{\partial g_w}{\partial \chi_w}, \tag{S83}$$

where $V(\partial\varepsilon_\infty/\partial V)$ is given by Eq. (S30), $\partial\rho/\partial Z_w = -\rho/Z_w$, and $\partial g_w/\partial Z_w$ follows from differentiating Eq. (S22) through P_{ww} and P_{wc} (Eqs. (S45)–(S48)) at fixed χ_w, χ_c, using Eq. (S75) for $\partial\delta_w/\partial Z_w$. Similarly, $\partial g_w/\partial\chi_w$ uses $\partial\chi_c/\partial\chi_w$ derived above.

Row 0: $\partial F_0/\partial \mathbf{v}$

The Debye screening length $\kappa = \left(e^2 N_A \rho \sum_j x_j z_j^2 / k_B T \varepsilon_0 \varepsilon_r\right)^{1/2}$ gives

$$\frac{\partial \kappa}{\partial Z_w} = -\frac{\kappa}{2Z_w}, \qquad \frac{\partial \kappa}{\partial \varepsilon_r} = -\frac{\kappa}{2\varepsilon_r}. \tag{S84}$$

For the DH contribution:

$$J_{01} = -\frac{\partial Z^{\mathrm{DH}}}{\partial \varepsilon_r} - \frac{\partial Z^{\mathrm{Perm}}}{\partial \varepsilon_r}. \tag{S85}$$

The first term follows from differentiating Eq. (S8) through $\kappa(\varepsilon_r)$ using Eq. (S84). The second requires differentiating $Z^{\mathrm{Perm}} = -a_{DH+B,\varepsilon_r}\, V(\partial\varepsilon_r/\partial V)$ with respect to ε_r, yielding

$$\frac{\partial Z^{\mathrm{Perm}}}{\partial \varepsilon_r} = -\left[\frac{\partial a_{DH+B,\varepsilon_r}}{\partial \varepsilon_r}\, V\left(\frac{\partial \varepsilon_r}{\partial V}\right) + a_{DH+B,\varepsilon_r}\, \frac{\partial}{\partial \varepsilon_r} V\left(\frac{\partial \varepsilon_r}{\partial V}\right)\right], \tag{S86}$$

where

$$\frac{\partial a_{DH+B,\varepsilon_r}}{\partial \varepsilon_r} = \frac{a_{DH+B,\varepsilon_r}}{H_{DH}}\, \frac{\partial H_{DH}}{\partial \kappa}\, \frac{\partial \kappa}{\partial \varepsilon_r} - \frac{2\, a_{DH+B,\varepsilon_r}}{\varepsilon_r}, \tag{S87}$$

with $H_{DH} = \sum_j x_j z_j^2(\kappa/(1+\kappa\sigma_j) - 1/R_{B,j})$ and $\partial H_{DH}/\partial\kappa = \sum_j x_j z_j^2/(1+\kappa\sigma_j)^2$. The derivative of $V(\partial\varepsilon_r/\partial V)$ with respect to ε_r is obtained by differentiating Eq. (S26) through

$\partial F/\partial \varepsilon_r$ and $\partial F/\partial \varepsilon_\infty$ (which both depend on ε_r):

$$\frac{\partial}{\partial \varepsilon_r} V\left(\frac{\partial \varepsilon_r}{\partial V}\right) = -\frac{V(\partial \varepsilon_r/\partial V)}{\partial F/\partial \varepsilon_r}\,\frac{\partial^2 F}{\partial \varepsilon_r^2} - \frac{1}{\partial F/\partial \varepsilon_r}\,\frac{\partial^2 F}{\partial \varepsilon_r\,\partial \varepsilon_\infty}\,V\left(\frac{\partial \varepsilon_\infty}{\partial V}\right), \tag{S88}$$

where the second-order partial derivatives of F (Eq. (S21)) are

$$\frac{\partial^2 F}{\partial \varepsilon_r^2} = -\frac{2\varepsilon_\infty^2}{\varepsilon_r^3(\varepsilon_\infty+2)^2}, \qquad \frac{\partial^2 F}{\partial \varepsilon_r\,\partial \varepsilon_\infty} = \frac{4(\varepsilon_\infty - \varepsilon_r^2)}{\varepsilon_r^2(\varepsilon_\infty+2)^3}. \tag{S89}$$

The entry J_{02} arises from the χ_w-dependence of Z^{assoc} and of Z^{Perm} (through $V(\partial \varepsilon_r/\partial V)$, which depends on g_w and thus on χ_w):

$$J_{02} = -\frac{\partial Z^{\text{assoc}}}{\partial \chi_w} - \frac{\partial Z^{\text{Perm}}}{\partial \chi_w}, \tag{S90}$$

where $\partial Z^{\text{assoc}}/\partial \chi_w$ follows directly from Eq. (S6) (differentiating through $\chi_c(\chi_w)$), and $\partial Z^{\text{Perm}}/\partial \chi_w = a_{DH+B,\varepsilon_r}\,(\partial V(\partial \varepsilon_r/\partial V)/\partial \chi_w)$ requires differentiating the chi cross-derivative system (Eqs. (S53) and (S54)) with respect to χ_w to obtain $\partial(V\partial \chi_w/\partial V)/\partial \chi_w$ and $\partial(V\partial \chi_c/\partial V)/\partial \chi_w$, and then propagating through P_{ww}, P_{wc}, g_w, and $V(\partial F/\partial V)$.

The entry J_{00} approximates $\partial(Z^{\text{phys}} + Z^{\text{assoc}} + Z^{\text{DH}} + Z^{\text{Perm}})/\partial Z_w$ by omitting one chain-rule contribution: the term arising from the implicit Z_w-dependence of $V(\partial \chi_w/\partial V)$ and $V(\partial \chi_c/\partial V)$ through the chi cross-derivative system. Validating against centered finite differences shows this residual to be 1–2 % at typical conditions, which does not impair Newton convergence from warm starts.

Parameters

Table S1: Pure component parameters in the CPA and eCPA EoS.[?] [?] [a]

Component	a_{0i}	c_{1i}	b_i	ϵ_i/k_B	κ_i	σ_i	R_{Bi}	μ_{0i}	α_{0i}	γ_{ii}	θ_{ii}	Υ_i
H_2O	0.1228	0.6736	14.52	2003	1.004	-	-	1.855	1.613	63.5	94.8	-
CO_2	0.3508	0.7602	27.20	-	-	-	-	-	2.695	-	-	-
Na^+	0	0	16.49	-	-	2.36	1.67	-	2.221	-	-	-53.5
Cl^-	0	0	40.83	-	-	3.19	1.83	-	3.557	-	-	

[a] Units: a_{0i} = [J.m^3.mol^{-2}]; c_{1i} = [-]; b_i = [cm^3.mol^{-1}]; ϵ_i/k_B = [K]; κ_i = [cm^3.mol^{-1}]; σ_i = [Å]; R_{Bi} = [Å] μ_{0i} = [D]; α_{0i} = [10^{40} C.m^{-2}.J^{-1}]; γ_{ii} = [º]; θ_{ii} = [º]; Υ_i = = [cm^3.mol^{-1}].

Table S2: LJ parameters and partial charge of atoms in H_2O, CO_2, and NaCl molecules from the respective force fields SPCE,[?] EPM2[?] and **?**.[?]

Atoms	σ_{ij} (Å)	ε_{ij} (kJ.mol^{-1})	q_i
O_w	3.166	0.65	-0.8476
H_w	0	0	+0.4238
C_c	2.757	0.2339	+0.6512
O_c	3.033	0.6694	-0.3256
Na^+	2.35	0.5439	+1
Cl^-	4.40	0.4184	-1

Table S3: Unlike LJ parameters between CO_2 and H_2O from Lorentz-Berthelot (LB) combining rules and optimization.[?] The number within parenthesis represents the ratio between OPT and LB parameters.

Atoms	LB		OPT	
	σ_{ij} (Å)	ε_{ij} (kJ·mol^{-1})	σ_{ij} (Å)	ε_{ij} (kJ·mol^{-1})
C_c-O_w	2.96	0.39	2.84 (0.96)	0.55 (1.41)
O_c-O_w	3.10	0.66	3.15 (1.02)	0.75 (1.14)

CO_2-Water Mixture

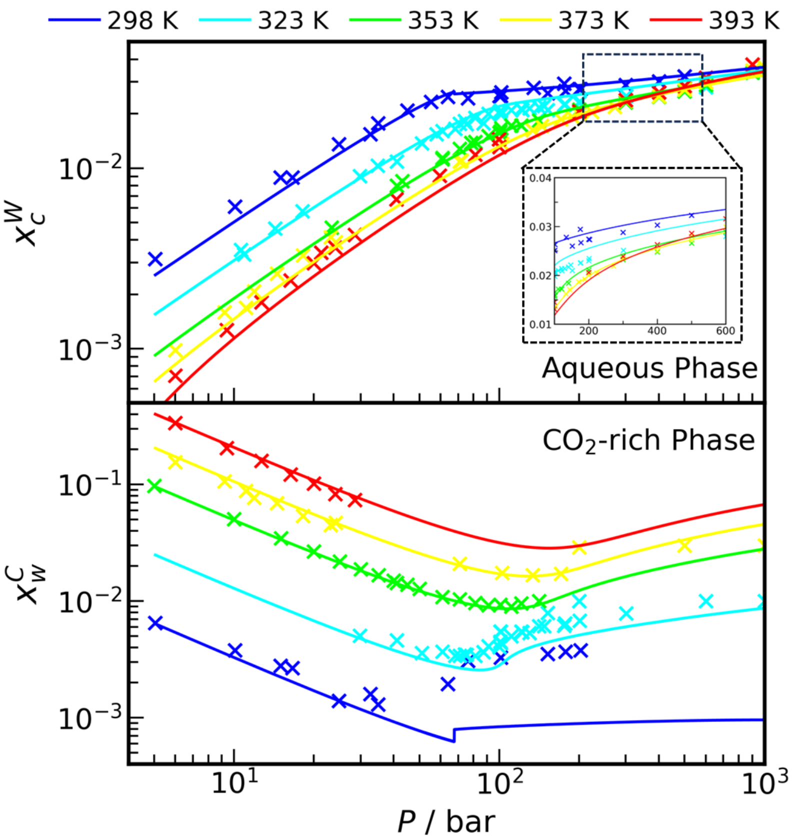


Figure S1: CO_2 solubility in the aqueous phase (upper plot) and water solubility in the CO_2-rich phase (lower plot) *vs.* pressure at low temperature (T < 400 K) in the CO_2-water mixture from CPA EoS (continuous line) and experiments[?,?,?,?,?,?,?,?,?,?,?,?,?,?] (discrete points).

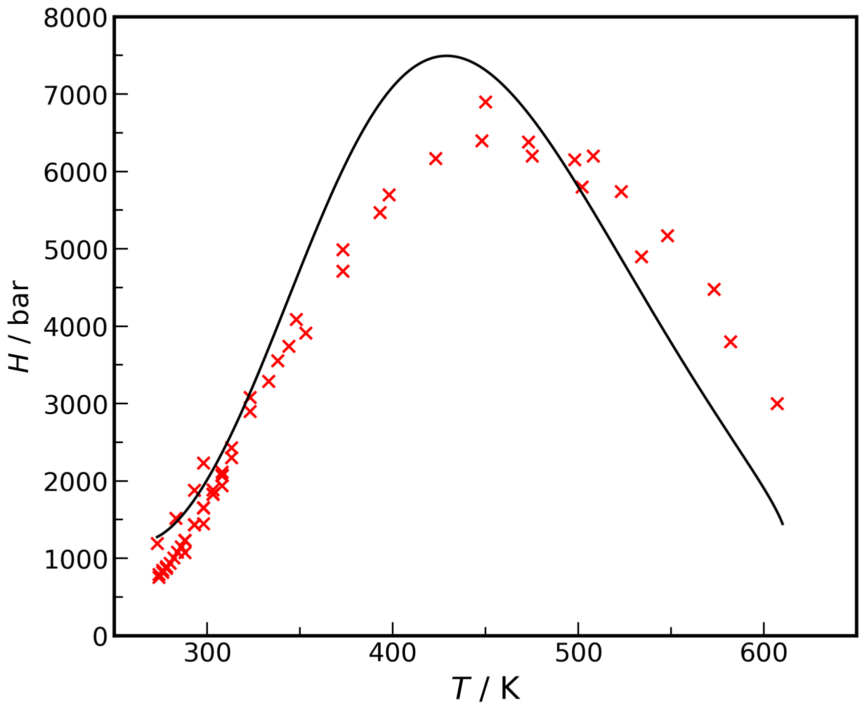
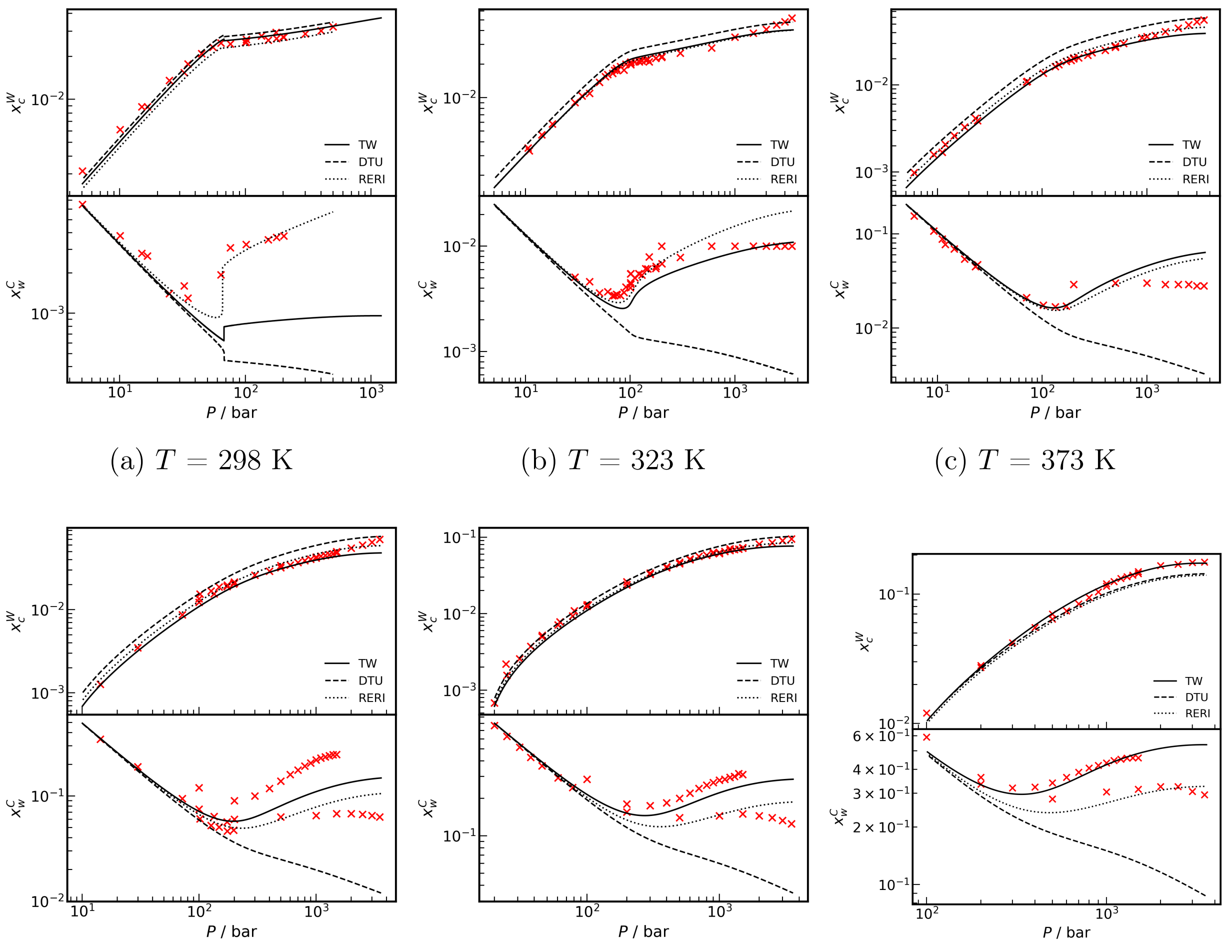


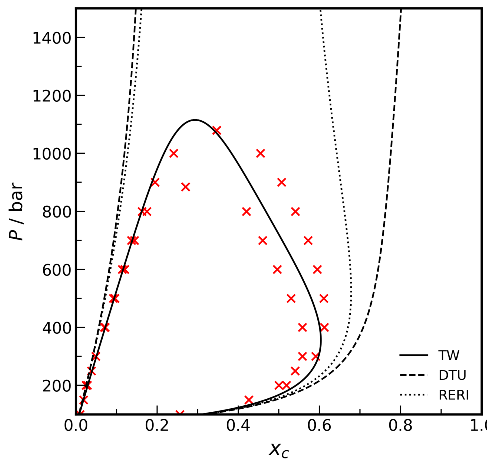
Figure S2: CO_2 Henry's constant (H) in water *vs.* temperature from CPA EoS (continuous line) and experiments[? ? ? ? ? ? ? ? ? ? ?] (discrete points).



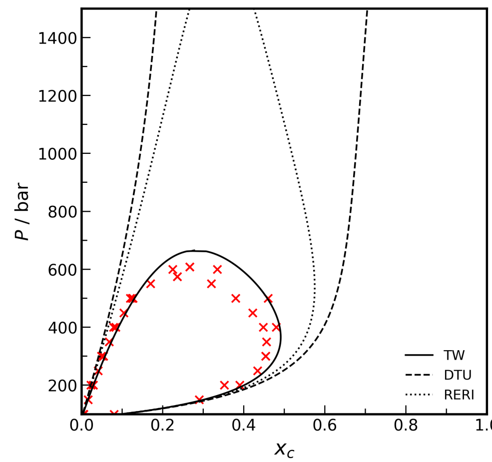
(a) T = 298 K (b) T = 323 K (c) T = 373 K

(d) T = 423 K (e) T = 473 K (f) T = 523 K

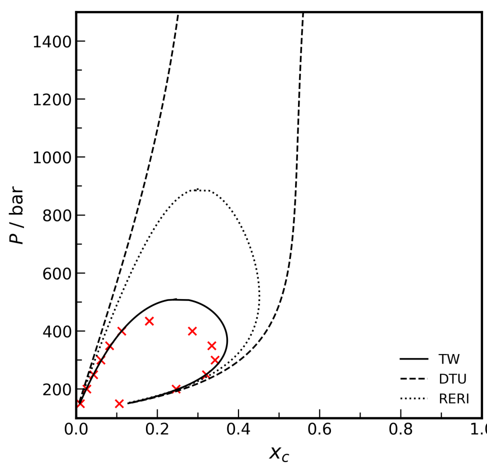
Figure S3: CO_2 solubility in the aqueous phase (upper plot) and water solubility in the CO_2-rich phase (lower plot) *vs.* pressure at various temperatures (a-f) from CPA EoS models (This Work (TW), DTU? ? ? and RERI?) and experiments? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? (discrete points).



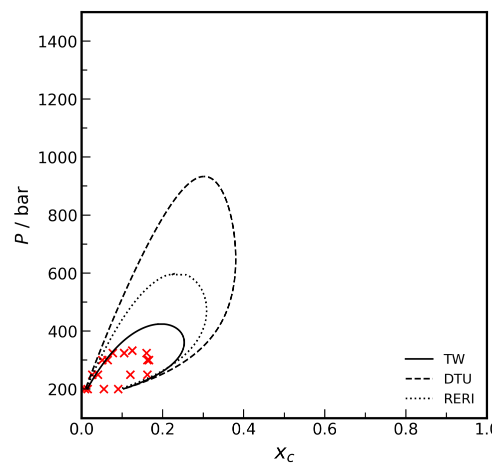
(a) $T = 548$ K



(b) $T = 573$ K

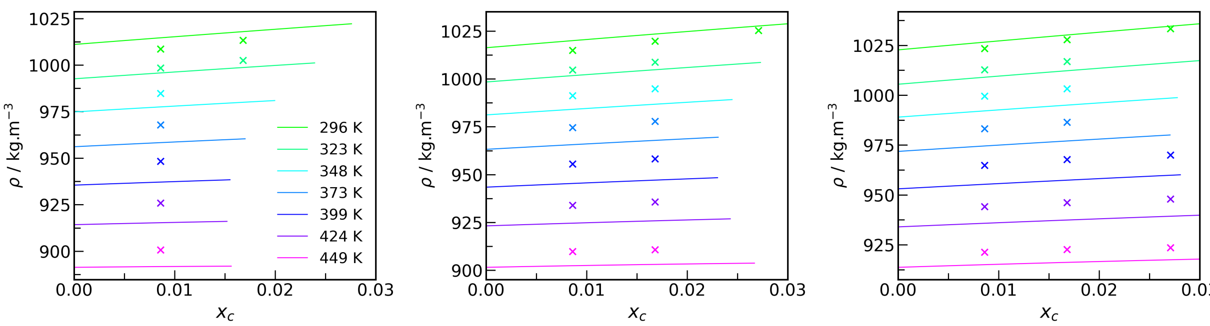


(c) $T = 598$ K

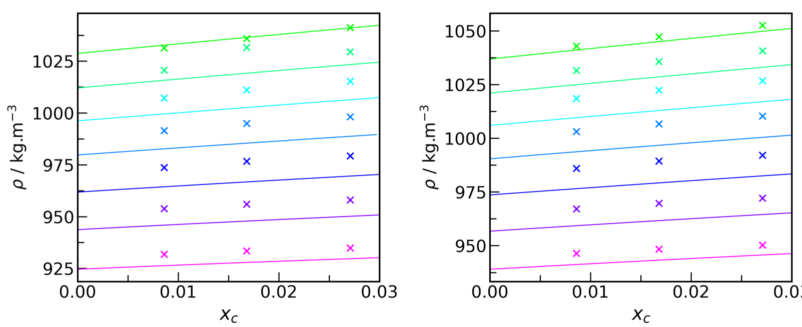


(d) $T = 623$ K

Figure S4: Phase diagram of the CO_2-H_2O mixture at various temperatures (a-d) from CPA EoS models (This Work (TW), DTU[?][?][?] and RERI[?]) and experiments[?][?][?] (discrete points).



(a) P = 150 bar (b) P = 300 bar (c) P = 500 bar



(d) P = 700 bar (e) P = 1000 bar

Figure S5: Density of CO_2-water mixture in the aqueous phase below the saturation point *vs.* CO_2 concentration at various temperatures (the legend in the subplot (a) holds for all others) and various pressures (a-e) from CPA EoS (continuous line) and experiments[?] (discrete points).

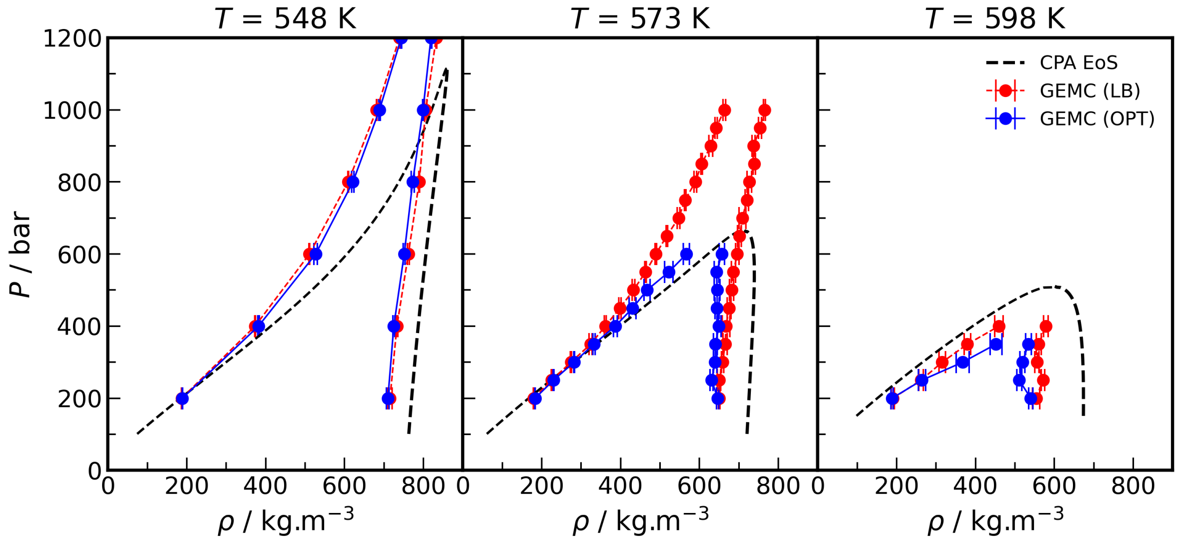


Figure S6: Density of the aqueous and CO_2-rich phases at equilibrium in the CO_2-H_2O mixture at various temperatures from GEMC simulations and CPA EoS. "LB" and "OPT" stand for Lorentz-Berthelot and optimized (at lower temperatures[?]) unlike parameters between CO_2-H_2O molecules, respectively.

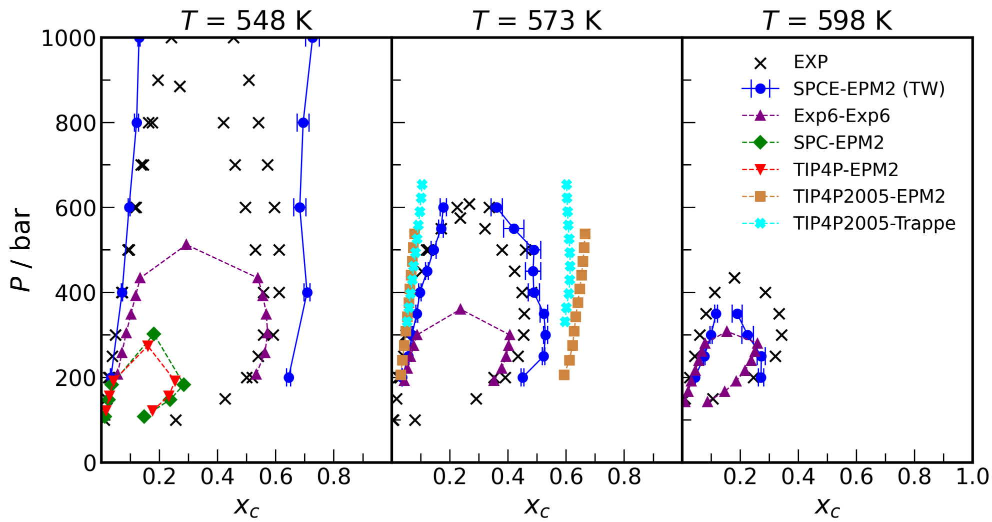


Figure S7: Phase diagram of the CO_2-H_2O mixture at various temperatures from GEMC simulations and experiments[?] [?] [?] for various classical force field combinations: SPCE-EPM2 (results from This Work using "OPT" unlike parameters), Exp6-Exp6,[?] SPC-EPM2,[?] TIP4P-EPM2,[?] TIP4P2005-EPM2,[?] TIP4P2005-Trappe.[?]

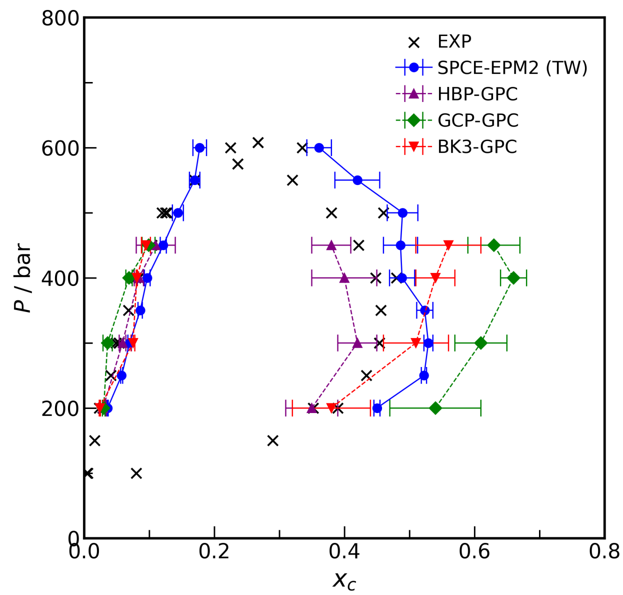


Figure S8: Phase diagram of the CO_2-H_2O mixture at T = 573 K from GEMC simulations and experiments[?] [?] [?] for various polarizable force field combinations: HBP-GPC,[?] GCP-GPC[?] and BK3-GPC.[?] We also show the results from the classical force field SPCE-EPM2 (results from This Work using "OPT" unlike parameters).

CO_2-Brine Mixture

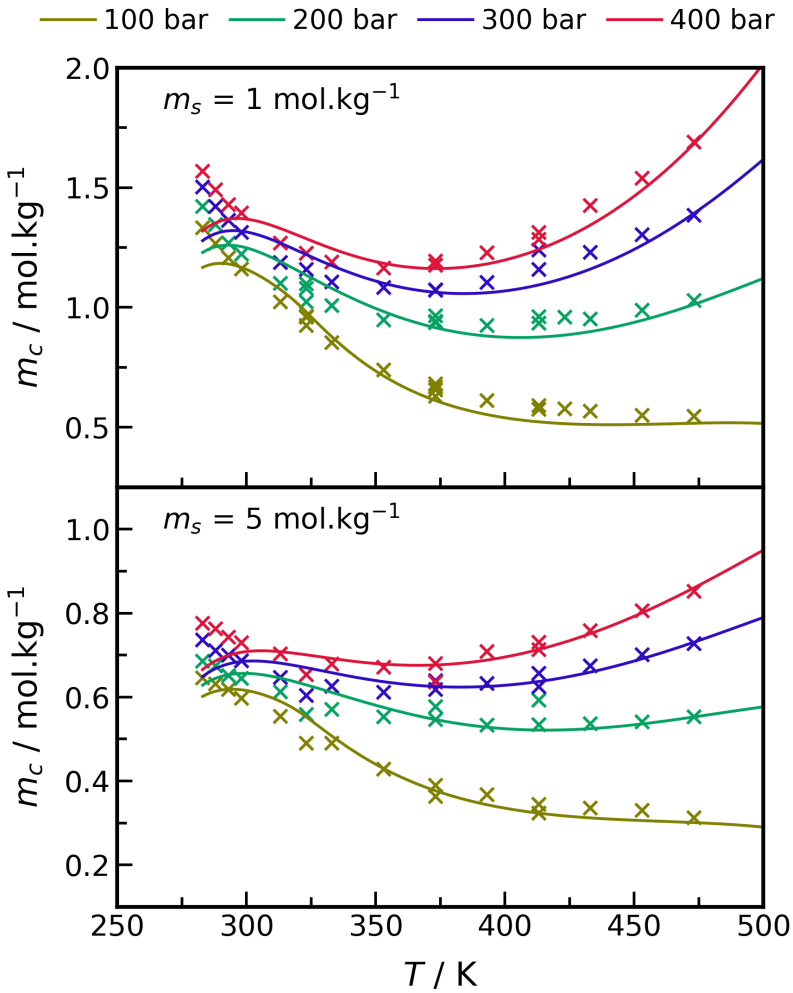
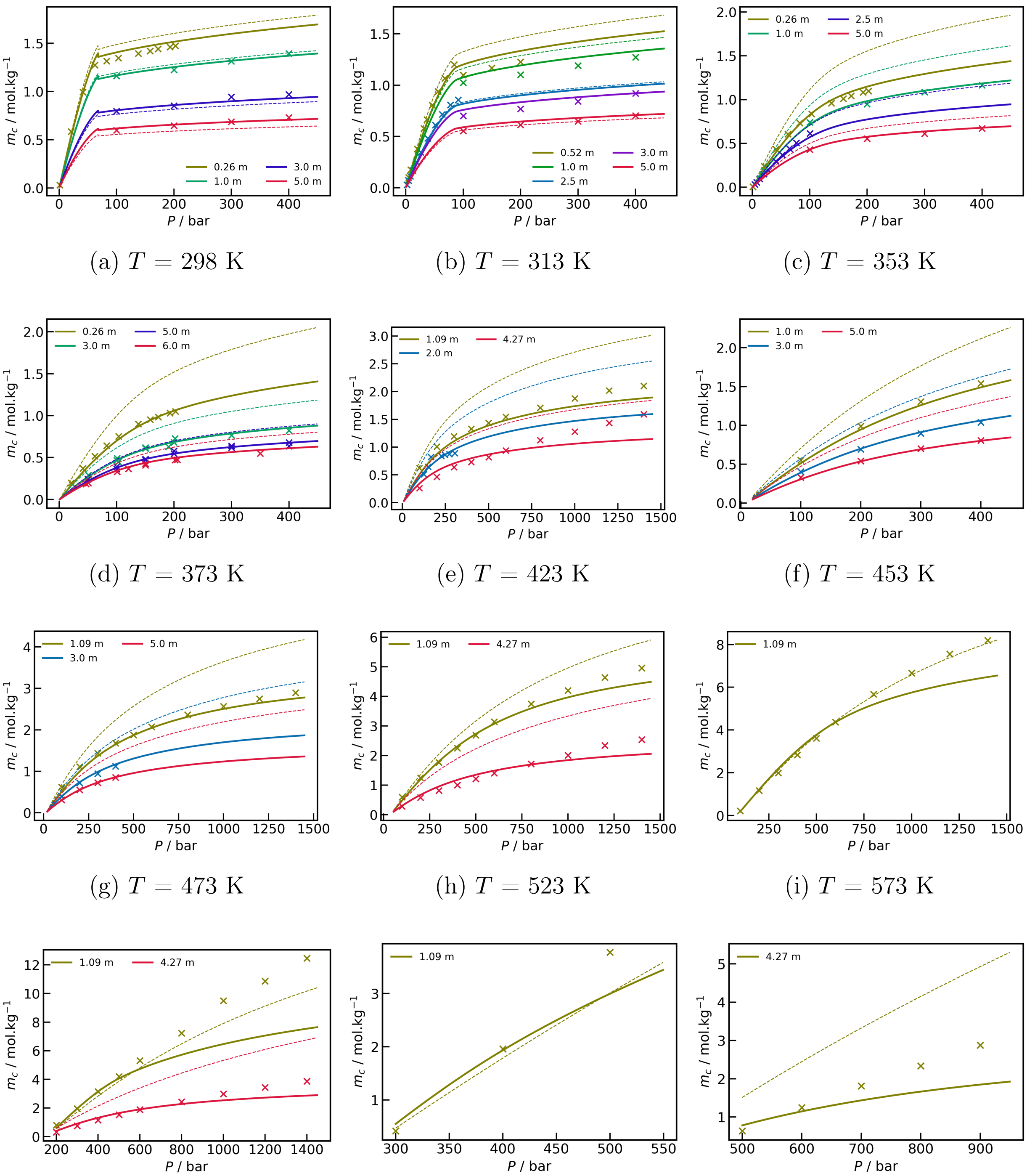


Figure S9: CO_2 solubility in NaCl brine from low to moderate temperatures ($T < 500$ K), various pressures and salinities of 1 and 5 mol.kg^{-1} from eCPA EoS (continuous line) and experiments[?][?][?][?] (discrete points).



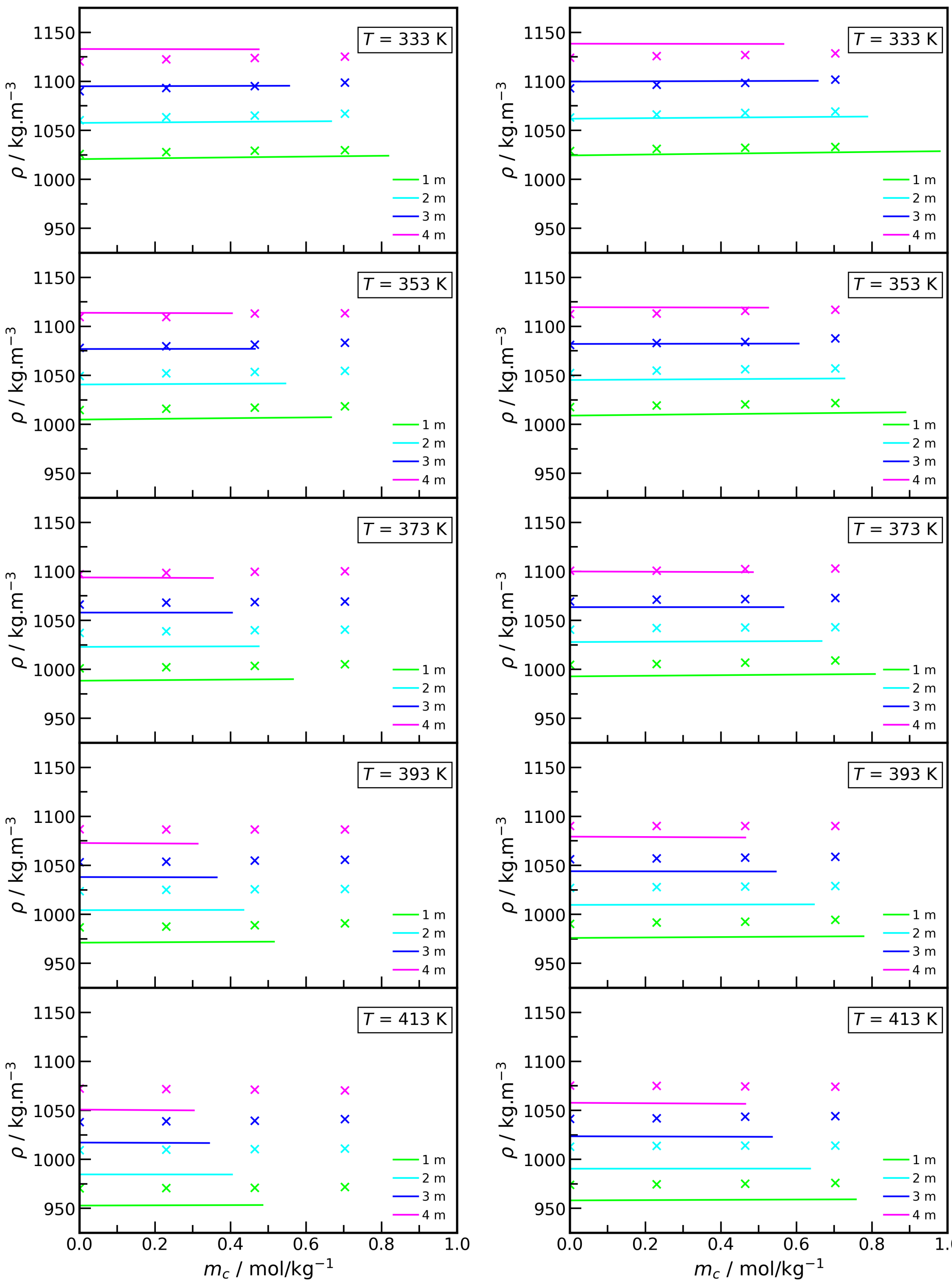
(a) T = 298 K (b) T = 313 K (c) T = 353 K

(d) T = 373 K (e) T = 423 K (f) T = 453 K

(g) T = 473 K (h) T = 523 K (i) T = 573 K

(j) T = 623 K (k) T = 673 K (l) T = 723 K

Figure S10: CO_2 solubility in NaCl brine at various temperatures (a-l) from eCPA models (This Work in continuous lines and DTU[?] in dashed lines) and experiments[?,?,?,?,?,?,?,?,?,?,?,?,?,?] (discrete points).



(a) P = 100 bar (b) P = 180 bar

Figure S11: Density of the aqueous phase in the CO_2-brine mixture below CO_2 saturation *vs.* CO_2 concentration at various pressures, salinities and two different pressures (a-b) from eCPA EoS (continuous line) and experiments[?] (discrete points).

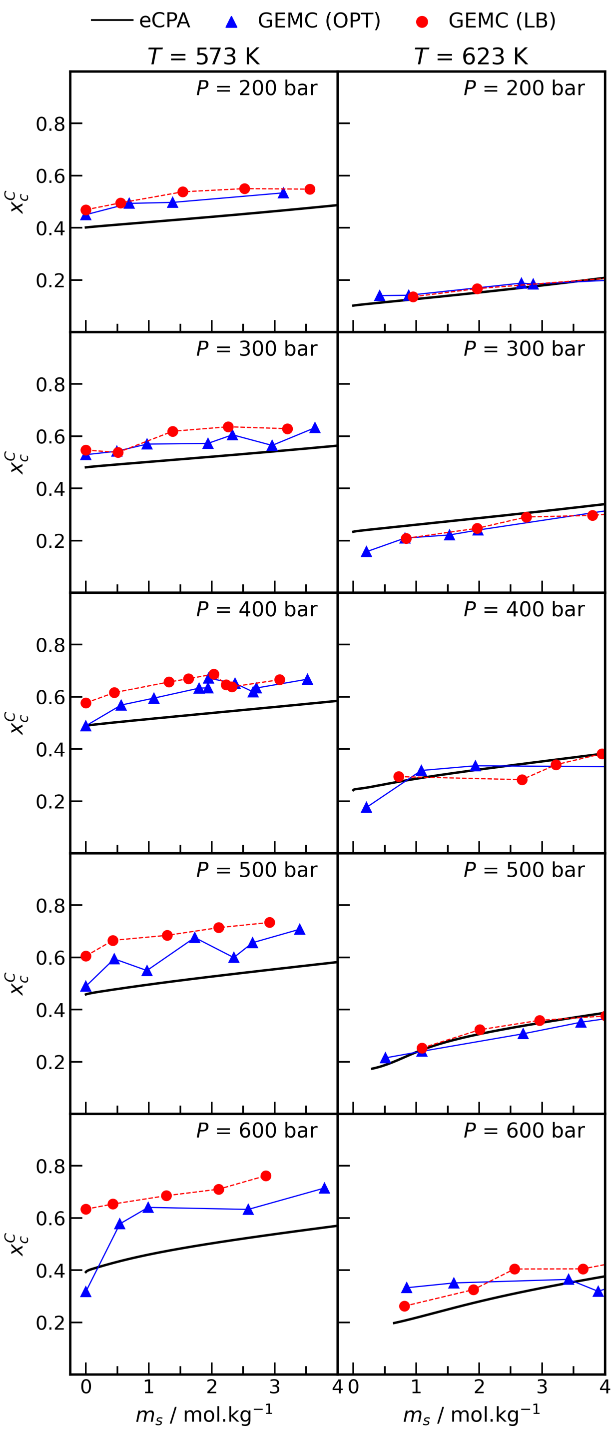


Figure S12: CO_2 solubility in the CO_2-rich phase in the CO_2-brine mixture *vs.* salt concentration at various pressures and two different temperatures from GEMC simulations and eCPA EoS. "LB" and "OPT" stand for Lorentz-Berthelot and optimized (at lower temperatures?) unlike parameters between CO_2-H_2O molecules, respectively.

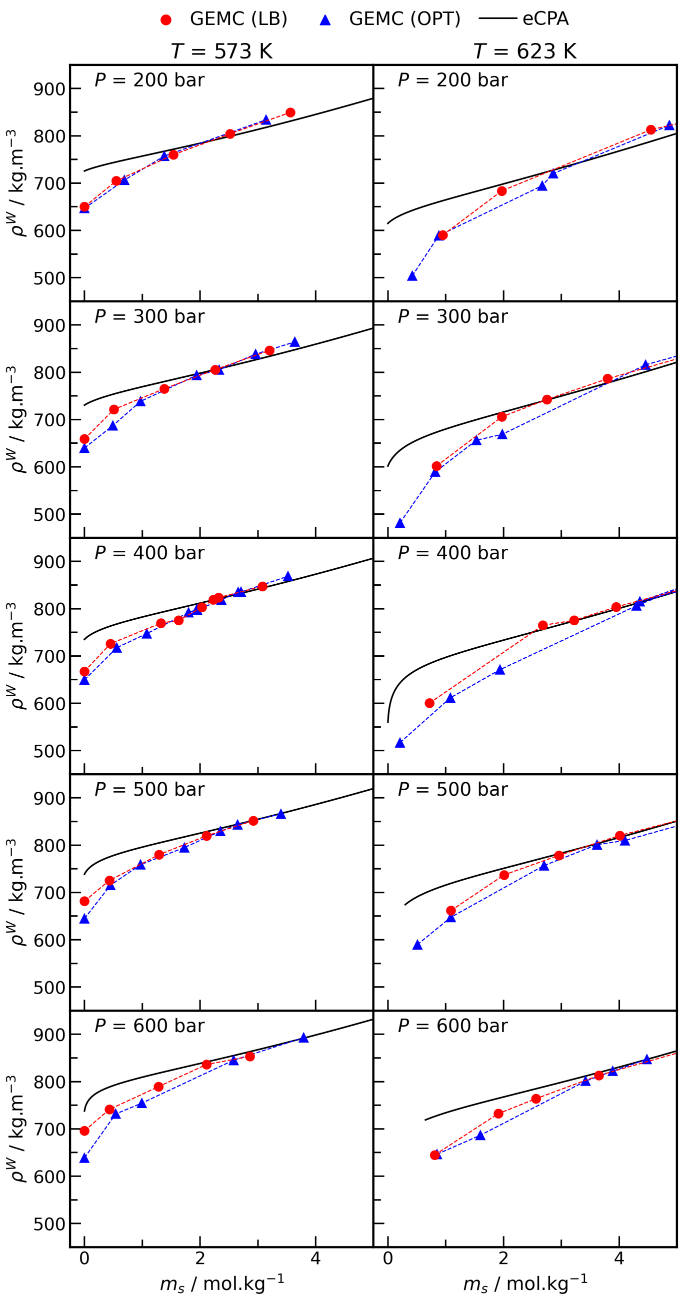


Figure S13: Density of the CO_2-saturated NaCl brine *vs.* salt concentration at various pressures and two different temperatures from GEMC simulations and eCPA EoS. "LB" and "OPT" stand for Lorentz-Berthelot and optimized (at lower temperatures[?]) unlike parameters between CO_2-H_2O molecules, respectively.

Association

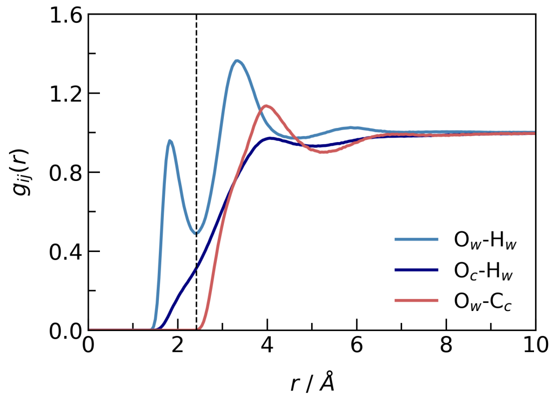
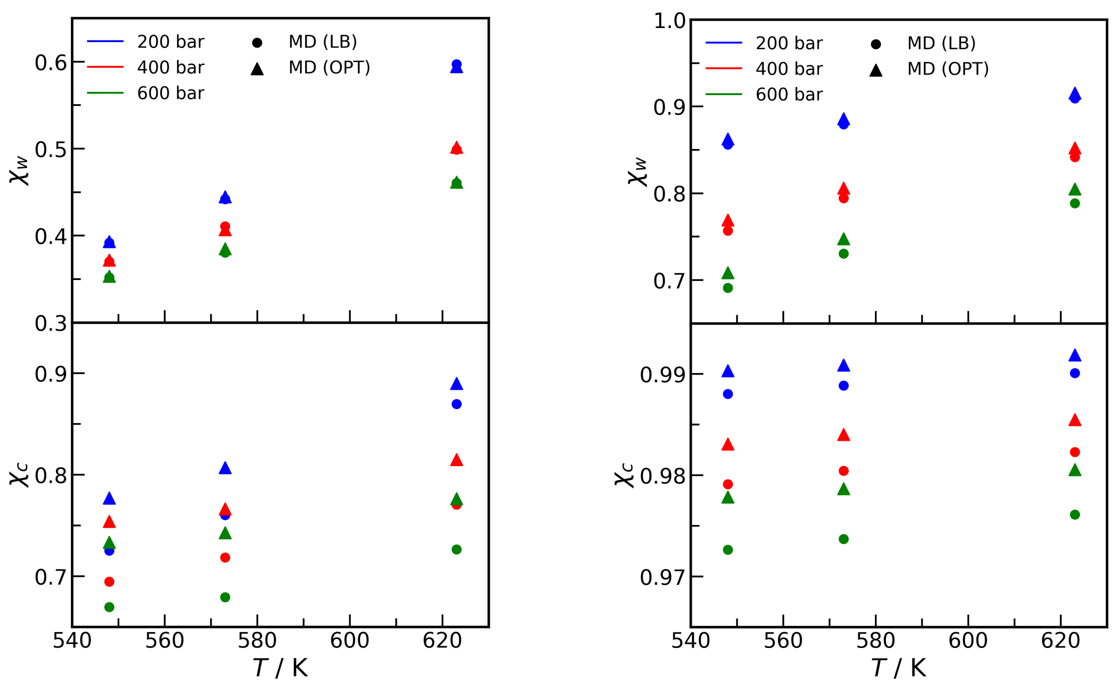


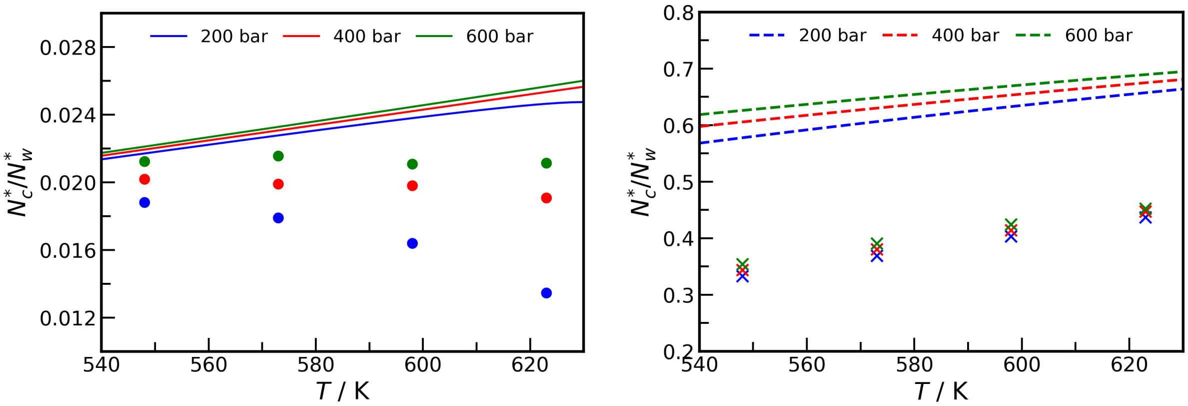
Figure S14: Scheme of the radial distribution function between the water-oxygen and water hydrogen, CO_2-oxygen and water-hydrogen, and water-oxygen and CO_2-carbon. The vertical dashed line represents the cutoff for the hydrogen bond criteria ($r = 2.4$ Å).



(a) $x_c = 0.04$

(b) $x_c = 0.80$

Figure S15: Association (χ_w) and cross-association (χ_c) in the CO_2-water mixture *vs.* temperature at various pressures in (a) aqueous phase ($x_c = 0.04$) and (b) CO_2-rich phase ($x_c = 0.80$) from MD simulations. "LB" and "OPT" stand for Lorentz-Berthelot and optimized (at lower temperatures[?]) unlike parameters between CO_2-H_2O molecules, respectively.



(a) $x_c = 0.04$ (b) $x_c = 0.80$

Figure S16: Association ratio between the number of associating CO_2 (N_c^*) and the number of associating H_2O (N_w^*) from MD simulations (discrete points) and CPA EoS (continuous line) in (a) aqueous phase ($x_c = 0.04$) and (b) CO_2-rich phase ($x_c = 0.80$).